Delamination & Popcorn Effect Analysis in IC Packaging

1. Problem Statement

If I observe failures during reliability qualification testing, the first thing I will focus on is when exactly the failure is triggered.

- In this scenario, I would notice a clear pattern:
  All units pass the complete assembly process successfully
- Failures appear only after thermal exposure during reliability testing, which simulates reflow conditions

What Makes This Important:

Reflow is a high-temperature process (~260°C) used in semiconductor packaging and PCB assembly.

During reliability testing, a similar thermal profile is applied to check whether the package can withstand real-world conditions.

So if failures occur after this thermal exposure, it indicates:

The package is structurally stable at room conditions, but fails under high temperature stress.

2. Initial Observations

After identifying that failures occur during reliability testing under reflow-like thermal exposure, I will begin by carefully observing the failure pattern.

Observation 1: Stage of Failure

I would first check at which stage the failure is introduced:

- Units pass all assembly stages successfully

- Failures are observed only after thermal exposure (reflow simulation)

This indicates:

The failure is not introduced during assembly, but is triggered under thermal stress

Observation 2: Nature of Failure

Next, I would observe the type of defects:

- Package cracking
- Internal delamination
- In some cases, electrical failures (likely secondary)

This suggests:

The failure is primarily mechanical in nature

Observation 3: Pattern of Occurrence

I would check whether the failures are random or patterned:

- Failures increase with higher thermal exposure or environmental stress
- Not random across all units

This indicates:

The issue is systematic, not random variation

Observation 4: Possible Environmental Link

Since failures appear after thermal exposure, I would suspect:

- Moisture presence inside the package
- Interaction between temperature and internal material behavior

Key Insight from Observations

From these observations, I can conclude:

The failure is thermally triggered, mechanical in nature, and likely influenced by environmental conditions such as moisture.

3. Creating & Analyzing Dataset

After identifying that the failure is thermally triggered during reflow-like exposure, I will move to data analysis to validate my observations.

Since real production data is not available, I will create a simulated production-like dataset that mimics real manufacturing conditions.

Why is a dataset needed?

At this stage, I want to answer:

Is there a measurable factor that correlates with the failure?

From observations, moisture is a strong suspect.

So I will design the dataset to study:

Moisture exposure vs Failure rate

<u>How This Data is Obtained in Real Production</u>

In an actual OSAT manufacturing environment, this type of dataset would be obtained from multiple sources across the production and reliability flow.

Data Sources

I would collect data from:

- ATE (Automated Test Equipment)

  → Provides pass/fail results and failure classification after testing

- Reliability Testing Reports

  → Includes results from reflow simulation and moisture sensitivity testing (MSL)

- MES (Manufacturing Execution System)

  → Tracks lot-wise production history, including process conditions and timestamps

Lot	Moisture Exposure Level	Units Tested	Failures	Failure %
L1	Low	1000	5	0.50%
L2	Low	1000	8	0.80%
L3	Medium	1000	25	2.50%
L4	High	1000	60	6.00%
L5	High	1000	65	6.50%

Observation from Data

Now I analyze the trend:

- Low moisture → very low failure
- Medium moisture → noticeable increase
- High moisture → sharp increase in failures

This shows a clear pattern:

Failure rate increases with moisture exposure

Key Insight:

From this dataset, I can conclude:

There is a strong correlation between moisture exposure and failure during thermal (reflow-like) conditions

Engineering Thinking

At this point, I am not yet confirming the root cause.

But I am narrowing it down:

- Moisture is likely playing a critical role

- Thermal exposure (reflow) acts as a trigger
- The failure mechanism is likely related to internal pressure or stress

Why This Step is Important

This step proves:

- I use data, not assumptions
- I validate observations
- I identify patterns before concluding

4. Understanding the Failure Type

After analyzing the dataset and observing the failure pattern, I will now focus on understanding what type of failure is occurring.

Step 1: Identify Failure Nature

From initial observations, I would note:

- Package cracking
- Internal delamination
- Occasional electrical failures

This immediately tells me:

The primary failure is mechanical, not purely electrical

Step 2: Link Mechanical to Electrical (Important)

Even though some units show electrical failure, I would think:

- Electrical failure is likely a secondary effect
- Caused by:
    - Broken interconnect

    - Delaminated interfaces
    - Stress-induced damage

So I conclude:

Electrical failure is a result, not the root problem

Step 3: Where Can This Failure Occur?

Now I ask:

Where inside the package can this type of failure happen?

Possible locations:

- Between the molding compound and the die
- Between the molding compound and substrate
- Within the molding compound itself

This points toward:

Interface-related failure (delamination)

Step 4: What Type of Mechanical Failure?

Now I refine further:

- Cracks → stress-related
- Delamination → interface separation
- Occurs after heating → thermal-driven

So I define:

This is a thermally induced mechanical failure, involving delamination and cracking

Final Understanding:

The failure is primarily a mechanical failure caused by internal stress during thermal exposure, leading to delamination and package cracking, with possible secondary electrical failures.

Engineering Thinking

At this stage, I have:

- Identified failure type → Mechanical
- Identified nature → Thermally triggered
- Identified location → Internal interfaces

But still one question remains:

What is generating the internal stress?

5. Identifying Failure Stage

After understanding that the failure is mechanical and thermally induced, I will now determine at which stage the failure is actually introduced.

Step 1: Review Full Process Flow

I will first map the complete backend process:

- Wafer
- Die Attach
- Wire Bond / Flip-Chip
- Molding
- Ball Attach
- Reliability Testing (including reflow simulation)
- Final Test

Step 2: Correlate Failure Timing

From earlier observations:

- Units pass all assembly stages
- No visible defects before testing
- Failures appear only after thermal exposure during reliability testing (reflow simulation)

This is a critical clue:
The failure is not introduced during assembly but is triggered during thermal exposure

Step 3: Interpret What This Means

Now I think deeper:

- The assembly process is not directly defective
- But something inside the package is vulnerable
- That weakness becomes visible only when:
    - Temperature increases
    - Internal stress develops

So I conclude:
The failure mechanism is activated during reflow-like thermal conditions

Step 4: Stage Classification

Stage	**Role**
Assembly	Build the structure
Reliability Testing (Reflow Simulation)	**Triggers the failure**

Important distinction:

The root cause may originate earlier (e.g., moisture absorption during storage)
But the failure manifests during reflow thermal exposure

Final Conclusion

The failure is introduced as a latent condition during earlier stages (such as moisture absorption), but it is triggered and observed during reflow-equivalent thermal exposure in reliability testing.

Engineering Thinking:

At this point, I understand:

- Where failure is observed → After reflow
- Why it appears → Thermal stress
- What kind of issue → Latent + triggered failure

Now the key question becomes:

What exactly is inside the package that is causing this stress?

6. Physical Failure Mechanism

After identifying that the failure is triggered during reflow-like thermal exposure, I will now focus on understanding what physically happens inside the package.

Step 1: Identify the Source of Internal Stress

From earlier sections:

- Failure occurs only after thermal exposure
- Mechanical damage is observed (cracks, delamination)

So I ask:

What is inside the package that can generate stress when heated?

Step 2: Consider Material Behavior

Semiconductor packages are made of materials like:

- Molding compound (epoxy-based)
- Substrate
- Die

Important property:
These materials can absorb moisture from the environment

Step 3: What Happens During Thermal Exposure?

Now I think about reflow conditions (~260°C):

- Moisture inside the package gets heated
- Moisture turns into vapor
- Vapor expands rapidly

This leads to:
Internal pressure builds up inside the package

Step 4: Effect of Internal Pressure

This pressure has no easy escape path, so it creates stress at weak interfaces:

- Between the molding compound and die
- Between the molding compound and the substrate

As pressure increases:

- Interfaces start to separate → delamination

- If stress is high → package cracking

This Phenomenon is Known As:

Popcorn Effect

Named because the package can crack suddenly, similar to popcorn popping

Engineering Understanding

Now I have complete clarity:

- Source → Moisture
- Trigger → Thermal exposure (reflow)
- Effect → Internal pressure
- Result → Delamination + cracking

This answers the key question:
What is generating the internal stress?

7. Root Cause

Based on:

- Failure pattern observed after thermal exposure
- Strong correlation with moisture exposure
- Mechanical nature of failure (delamination and cracking)
- Understanding of material behavior during reflow

I conclude that:

Moisture absorbed inside the package vaporizes during reflow-like thermal exposure, generating internal pressure that leads to delamination and package cracking (popcorn effect).

Supporting Logic

- Moisture → present before heating
- Reflow thermal exposure → converts moisture to vapor
- Vapor expansion → generates internal pressure
- Pressure → causes interface separation and cracking

Key Insight

The failure is not due to a defect in assembly, but due to the moisture sensitivity of the package, combined with thermal exposure conditions

Engineering Interpretation

- Root cause type → Material + environmental interaction
- Failure trigger → Thermal (reflow) exposure
- Failure mode → Delamination and cracking

8. Root Cause Confirmation & Validation

After identifying the moisture-induced popcorn effect as the root cause, I will not directly finalize it.

Instead, I will validate it through data correlation, inspection, and controlled experiments.

Step 1: Correlate with Moisture Conditions

I will compare failure data across different moisture exposure conditions:

- Low exposure → low failure

- High exposure → high failure

If this trend is consistent:

It strongly supports that moisture is influencing the failure

Step 2: Physical Inspection

I will use inspection techniques to verify internal damage:

- CSAM (C-Scan Acoustic Microscopy)

  → Detect internal delamination

- Visual inspection

  → Identify package cracks

Expected observation:

- Delamination at interfaces
- Crack propagation from internal regions

Expected Result:

- Set A (moisture present) → high failure
- Set B (moisture removed) → significantly reduced failure

If this happens:

Root cause is confirmed as moisture-driven failure

Step 4: Process Monitoring

I will also check:

- Moisture exposure duration
- MSL (Moisture Sensitivity Level) compliance
- Storage conditions

Any deviation here further supports the root cause

Final Validation Logic

If removing moisture reduces failure, then moisture is the root cause

9. Corrective Actions

After confirming that the root cause is moisture-induced failure during reflow-like thermal exposure, I will focus on implementing corrective actions to eliminate or reduce moisture inside the package.

Step 1: Moisture Control (Primary Action)

Since moisture is the root cause, the priority is to control it.

I will ensure:

- Proper MSL (Moisture Sensitivity Level) handling
- Limited exposure to ambient humidity
- Use of dry storage conditions (dry cabinets, moisture barrier bags)

Step 2: Pre-Bake Process

To remove absorbed moisture before thermal exposure:

I will introduce or enforce:

- Pre-baking of components before reflow

- Controlled baking temperature and duration

This ensures:
Moisture is removed before high-temperature exposure

Step 3: Process Control During Handling

I will ensure:

- Minimal delay between unpacking and reflow
- Proper tracking of exposure time
- Handling as per MSL guidelines

Step 4: Material / Package Improvement (If Needed)

If the issue persists:

I will evaluate:

- Use of low-moisture absorption molding compounds
- Improved package design to reduce internal stress

Expected Outcome

After implementing these corrective actions:

- Moisture content inside the package reduces
- Internal pressure during reflow is minimized
- Delamination and cracking are significantly reduced

Final Result (Expected)
Failure rate reduces from high levels (~6%) to normal acceptable levels (~1% or lower)

10. Final Result & Learning

After implementing the corrective actions and validating the root cause, I would expect a significant improvement in package reliability.

Final Result (Expected)

- Failure rate reduces from higher levels (~6%) to normal levels (~1% or lower)
- No recurrence of cracking or delamination under reflow-like thermal exposure
- Improved stability across production lots

Key Outcome

The reliability issue is resolved by controlling moisture and improving process handling, ensuring the package can withstand thermal exposure conditions.

Key Learnings:

Through this analysis, I gained a deeper understanding of:

- The impact of moisture on semiconductor package reliability
- How thermal exposure (reflow) can trigger internal stress
- The importance of MSL control and proper handling
- The concept of latent defects triggered under stress conditions
- How to approach problems using:
    - Observation
    - Data analysis
    - Physical understanding
    - Validation

Engineering Takeaway

Not all failures originate during manufacturing — some are latent and only appear under stress conditions like thermal exposure.

Final Thought

This case reinforces that:

Understanding material behavior + process conditions + environment is critical for solving real-world semiconductor reliability problems.

References

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